



# 3D Knitting

|                     |                                                                                                                                                                                                                                                                                       |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 🕒 Created by        | © Gustavo Vieira                                                                                                                                                                                                                                                                      |
| 🕒 Created time      | April 23, 2026 12:02 AM                                                                                                                                                                                                                                                               |
| ☰ Category          | Methodology doc                                                                                                                                                                                                                                                                       |
| 🕒 Last edited by    | © Gustavo Vieira                                                                                                                                                                                                                                                                      |
| 🕒 Last updated time | April 23, 2026 12:07 AM                                                                                                                                                                                                                                                               |
| 👤 Reviewers         |  Laurent  Vincent  Kishore Ganesan |

## 1. Introduction

### 1.1 Purpose of the study

This document describes the life cycle inventory (LCI) model for 3D / WHOLEGARMENT knitting as part of the Carbonfact Textile LCA Database. The aim is to provide a transparent, traceable, and updatable foreground inventory for 1 kg of 3D-knitted garment, consistent with the modelling conventions applied across all knitting technology documents in this database.

3D knitting machines — notably the Shima Seiki SWG-XR and MACH2XS series, and the Karl Mayer Stoll CMS 3D — integrate design, shaping, and knitting in a single pass, eliminating virtually all cutting and sewing waste. This distinctive characteristic gives 3D knitting its lowest yarn-loss rate in the knitting technology family, while also producing its highest specific energy consumption per kilogram of output, because complex garment geometries and slow knitting cycles increase machine time per unit mass.

### 1.2 System boundary

The system boundary covers gate-to-gate 3D knitting: from yarn input to finished 3D-knitted garment at the mill gate. The following flows are **included** within the foreground model:

- Direct electricity consumption of the knitting machine.
- Indirect energy: HVAC, compressed air, illumination, vacuum/suction for yarn handling, robotic take-down systems.
- Lubricant input and the corresponding waste mineral oil output (auxiliary waste — modelled inline).
- Capital goods: machine amortisation and EoL flows (auxiliary — modelled inline).
- Building infrastructure: allocated floor area per kg of garment.

The following are explicitly **excluded**:

- Fibre and yarn production (upstream).
- **Textile and material waste:** yarn losses are handled externally by the caller via the IOR. No inline yarn-loss or textile-waste exchange is included in the dataset itself.
- Finishing, packaging, and transport (downstream).

## 1.3 Functional unit

**1 kg of 3D knitted garment** at the knitting mill gate, ready for optional finishing.  
Background: ecoinvent 3.12, system model: cut-off by classification, GLO/RoW market activities.

---

## 2. Process Description

3D / WHOLEGARMENT knitting produces a complete, fully-fashioned three-dimensional garment on a flat-bed machine in a single operation, without any subsequent cut-and-sew steps. The reciprocating carriage traverses two opposing needle beds in a highly complex programme, forming shaped panels, sleeves, necklines, and pockets simultaneously. This eliminates 10–25% of fabric

waste typical of conventional cut-and-sew production (GIZ, 2023) and achieves material utilisation exceeding 95% (K&G Garment, 2025).

The price for this near-zero waste is energy intensity: garment cycle times are long (0.7–2.0 garments per hour depending on gauge and complexity), and the installed power of the machine (1.5–2.2 kW) is spread over a very low mass throughput — typically 0.3–0.5 kg per garment. The result is a specific electricity consumption substantially higher than in circular or flat panel knitting, with a published range of ~0.9 to over 17 kWh/kg depending on garment complexity, gauge, and machine speed (Shima Seiki datasheets; Knox, 2011; Rakib Ahmed, 2024).

The reference machines are the Shima Seiki SWG-XR124/XR154 (1,300–1,400 kg), Shima Seiki MACH2XS variants (1,000–1,212 kg), and the Karl Mayer Stoll CMS 3D (1,200 kg). A generic machine mass of **1,200 kg** is used for capital goods allocation, representing the mid-range of these specifications.

---

## 3. Electricity Model

### 3.1 Direct machine electricity

The direct electricity consumption adopted for the average scenario is **4.8 kWh/kg**. This value is derived from the mainstream operating scenario — 1.8 kW installed power at a productivity of 0.375 kg/h (approximately 1.2 garments per hour at 0.45 kg per garment per the Shima Seiki SWG-XR datasheet):

$$E = 1.8 \text{ kW} \div 0.375 \text{ kg/h} = 4.8 \text{ kWh/kg}$$

This is consistent with measurements reported in Knox (2011) for WHOLEGARMENT machines (4–8 kWh/kg for typical knitwear garments) and with the NTNU (2024) thesis measurements. The theoretical lower bound from the mainstream scenario (3.33 kWh/kg at 0.45 kg/h productivity) and the higher end of industry-reported values converge around 4–5 kWh/kg for average-complexity garments. The 4.8 kWh/kg value is adopted as the default pending primary factory data.

### 3.2 Indirect energy

**Shared methodology:** Indirect Energy in LCA — Shared Methodology (Knitting)  
— Full derivation, evidence base and subsystem distribution.

Indirect energy is modelled at **4.46 kWh/kg**, consistent with all knitting processes in this database. The derivation is:  $2.4 \div 0.35 \times 0.65 = 4.46$  kWh/kg (ADEME & Ministère de la Transition Écologique, 2024). In 3D knitting this category is particularly significant due to vacuum and suction systems for yarn handling, robotic take-down units, and the tight humidity control required for fine-gauge operation.

## 4. Life Cycle Inventory

The foreground inventory below covers the average scenario. Yarn losses and textile-waste treatment are excluded from the dataset and handled externally by the caller via the IOR (Section 5). Auxiliary waste flows — lubricant EoL and machine EoL — are modelled inline.

| Flow                                                  | Amount | Unit | Dataset                                               |                                        |
|-------------------------------------------------------|--------|------|-------------------------------------------------------|----------------------------------------|
| <b>Technosphere inputs</b>                            |        |      |                                                       |                                        |
| Lubricant, average                                    | 0.008  | kg   | `lubricating oil production {RoW} \                   | cut-off` — Sandin et al. (2019), p. 45 |
| Waste mineral oil (lubricant EoL)                     | 0.008  | kg   | `market for waste mineral oil {RoW} \                 | cut-off`                               |
| <b>Electricity</b>                                    |        |      |                                                       |                                        |
| Machine electricity (direct)                          | 4.8    | kWh  | `market group for electricity, medium voltage {GLO} \ | cut-off` — 1.8 kW ÷ 0.375 kg/h         |
| Indirect energy (HVAC, compressors, vacuum, lighting) | 4.46   | kWh  | `market group for electricity, medium voltage {GLO} \ | cut-off` — shared default              |

| Flow                                | Amount                 | Unit           | Dataset                                                                  |                                                  |
|-------------------------------------|------------------------|----------------|--------------------------------------------------------------------------|--------------------------------------------------|
| <b>Capital goods</b>                |                        |                |                                                                          |                                                  |
| Building machine (capital good)     | $1.667 \times 10^{-6}$ | unit           | `market for building machine {GLO} \                                     | cut-off` — 1,200 kg ÷ 360,000 kg ÷ 2,000 kg/unit |
| Building infrastructure             | $4.86 \times 10^{-4}$  | m <sup>2</sup> | `market for building, hall {GLO} \                                       | cut-off`                                         |
| <b>Waste flows (auxiliary only)</b> |                        |                |                                                                          |                                                  |
| EoL machine — steel (85%)           | $2.83 \times 10^{-3}$  | kg             | `market for waste steel {RoW} \                                          | cut-off` — 1,020 kg ÷ 360,000 kg                 |
| EoL machine — other (15%)           | $5.0 \times 10^{-4}$   | kg             | `treatment of waste electric and electronic equipment, shredding {GLO} \ | cut-off`                                         |

**Note:** the foreground dataset does not include yarn input or textile-waste flows. Yarn losses (2% default) are handled externally via the IOR by the caller.

## 5. Capital Goods and Building Infrastructure

**Shared methodology:** Capital Goods & End-of-Life (Machinery) in LCA — Full modelling framework and default values. This section states only 3D-knitting-specific inputs.

**Shared methodology:** Building Infrastructure in LCA — Full allocation framework. Section 5.2 states only the 3D-knitting-specific value.

Capital goods are included in accordance with EF guidance (Fazio et al., 2020) and ecoinvent practice (Weidema et al., 2013).

## 5.1 Machine parameters

The reference machine mass for 3D knitting is **1,200 kg**, based on the Karl Mayer Stoll CMS 3D catalogue value and the mid-range of the Shima Seiki SWG-XR and MACH2XS specifications (1,000–1,400 kg). Productivity is set at **5 kg/h**, derived from the mainstream scenario of ~1.2 garments/h at ~0.45 kg/garment on a mainstream operating cell.

| Parameter                         | Value                                    | Source / note                                                       |
|-----------------------------------|------------------------------------------|---------------------------------------------------------------------|
| Machine mass (kg)                 | 1,200                                    | Mid-range of Shima SWG-XR / MACH2XS / Stoll CMS 3D                  |
| Service life (years)              | 15                                       | Average scenario                                                    |
| Operating hours per day           | 16                                       | Average scenario                                                    |
| Operating days per year           | 300                                      | Average scenario                                                    |
| Productivity (kg/h)               | 5                                        | ~1.2 garments/h × 0.45 kg/garment × multi-machine cell; Knox (2011) |
| <b>Lifetime output Q (kg)</b>     | <b>360,000</b>                           | $15 \times 16 \times 300 \times 5$                                  |
| Machine allocation (kg/kg fabric) | 0.00333                                  | $1,200 \div 360,000$                                                |
| ecoinvent units (÷ 2000 kg/unit)  | <b><math>1.667 \times 10^{-6}</math></b> | $0.00333 \div 2,000$                                                |

Machine EoL: 85% steel (1,020 kg) →  **$2.83 \times 10^{-3}$  kg/kg fabric** → market for waste steel {RoW} | cut-off ; 15% other (180 kg) →  **$5.0 \times 10^{-4}$  kg/kg fabric** → treatment of waste electric and electronic equipment, shredding {GL0} | cut-off .

## 5.2 Building infrastructure

Building infrastructure is allocated using the Shima Seiki MACH2VS envelope (3,430 mm × 3,266 mm ≈ 11.2 m<sup>2</sup>) × cell factor ~3, yielding A<sub>alloc</sub> ≈ 34 m<sup>2</sup>. The conservative default adopted is  **$4.86 \times 10^{-4}$  m<sup>2</sup>/kg** — the highest building intensity in the knitting technology group, consistent with 3D knitting's substantially lower throughput per allocated space relative to circular and flat panel knitting.

ecoinvent activity: market for building, hall {GL0} | cut-off (unit: m<sup>2</sup>).

## 6. Input–Output Ratio (IOR)

**Definition:**  $\text{IOR} = 1 / (1 - L)$ , where  $L$  is the yarn loss rate (as a fraction). For every 1 kg of finished 3D knitted garment, the upstream yarn input must be scaled by IOR.

### How the loss rate is derived

3D / WHOLEGARMENT knitting produces complete fully fashioned garments with near-zero cutting waste, achieving >95% material utilisation (K&G Garment, 2025; GIZ, 2023). The **2% default loss rate** accounts for yarn breaks and machine setup trials — the lowest IOR in the knitting technology group. Sources: Advanced Knitting Technology (2021); Rakib Ahmed (2024, NTNU thesis).

### IOR value

| Scenario | Loss (%) | IOR (kg/kg) | Input required (kg) | Loss per kg output (kg) |
|----------|----------|-------------|---------------------|-------------------------|
| Default  | 2%       | 1.0204      | 1.0204              | 0.0204                  |

**How to apply:** for 1 kg of 3D knitted garment, multiply upstream yarn input by **1.0204**. The 0.0204 kg of lost yarn is routed by the caller to `market for waste yarn and waste textile {Row} | cut-off`. No inline yarn-loss or textile-waste exchange is included in the dataset itself.

## 7. Inferences and Limitations

The 4.8 kWh/kg machine electricity value carries the largest uncertainty in this inventory. Published values span nearly two orders of magnitude (0.9–17 kWh/kg) reflecting the strong dependence on garment complexity, gauge, and productivity. The mainstream scenario derivation ( $1.8 \text{ kW} \div 0.375 \text{ kg/h}$ ) provides a physically transparent anchor, but primary mill measurements from a representative sample of commercial 3D knitting production would substantially reduce this uncertainty. The indirect energy estimate (4.46 kWh/kg) is shared with all knitting technologies in this database. For 3D knitting, where vacuum systems and robotic take-down

units may represent a higher share of total plant electricity than in circular or flat knitting, sub-metered primary data are especially valuable.

Capital goods contribute less than 1% of total process GWP and are not a material source of uncertainty for most applications.

---

## 8. References

ADEME & Ministère de la Transition Écologique. (2024). *Ecobalyse methodological guide — Textiles and footwear* (Version 1.0). Paris, France.

Advanced Knitting Technology. (2021). *Circular and 3D knitting yarn realisation benchmarks*.

Fazio, S., Castellani, V., & Sala, S. (2020). *Guide for EF-compliant datasets*. European Commission, JRC.

[https://eplca.jrc.ec.europa.eu/permalink/Guide\\_EF\\_DATA.pdf](https://eplca.jrc.ec.europa.eu/permalink/Guide_EF_DATA.pdf)

GIZ. (2023). *Sustainable apparel report — Cutting waste reduction*. Deutsche Gesellschaft für Internationale Zusammenarbeit.

Hasanbeigi, A., & Price, L. (2015). A technical review of emerging technologies for energy and water efficiency in the textile industry. *Journal of Cleaner Production*, 95, 30–44. <https://doi.org/10.1016/j.jclepro.2015.02.079>

K&G Garment. (2025). *What is seamless knitting?*  
<https://kkgarment.com/blogs/news/what-is-seamless-knitting-a-complete-guide-for-brands>

Karl Mayer Stoll. (2025). *CMS 3D machine catalogue*. Karl Mayer.

Knox, J. (2011). *Knitted garment production energy analysis* (Unpublished thesis). Nottingham Trent University. <http://irep.ntu.ac.uk/id/eprint/18067/>

Laursen, S. E., Hansen, J., Knudsen, H. H., Wenzel, H., Larsen, H. F., & Kristensen, F. M. (2007). *EDIPTEx — Environmental assessment of textiles*. Danish Environmental Protection Agency.

Maity, S., Singha, K., & Singha, M. (2021). Advances in knitting technology. In *Advanced Knitting Technology* (pp. 225–260). Woodhead.  
<https://doi.org/10.1016/B978-0-12-821290-3.00009-0>



Rakib Ahmed. (2024). *Environmental impact of knitwear production: A gate-to-gate LCA* (Master's thesis). NTNU.

Sandin, G., Roos, S., & Johansson, M. (2019). *Environmental impact of textile fibres*. Mistra Future Fashion, IVL.

Shima Seiki. (2025). *WHOLEGARMENT SWG-XR / MACH2XS technical datasheets*. <https://www.shimaseiki.eu/>

Van der Velden, N. M., Patel, M. K., & Vogtländer, J. G. (2014). LCA benchmarking study on textiles. *Int. J. LCA*, 19(2), 331–356. <https://doi.org/10.1007/s11367-013-0626-9>

Weidema, B. P. et al. (2013). *Data quality guideline for ecoinvent v3*. <https://doi.org/10.25678/0006HH>